

Customer Shopping Behavior Analysis

End-to-end data analytics project — Python, SQL, Power BI & Gamma

Overview

A retail company wanted to understand what drives customer purchasing decisions and repeat business — factors like discounts, reviews, seasonality, and payment preferences — in order to improve sales, engagement, and long-term loyalty.

This project answers that question end to end:

- **Python** for data cleaning, transformation, and feature engineering
- **SQL (PostgreSQL)** for structured business-question analysis
- **Power BI** for an interactive dashboard
- **Gamma** for the stakeholder presentation
- A written report summarizing findings and recommendations

Dataset

- **Size:** 3,900 customer transactions, 18 columns
- **Fields:** customer demographics (age, gender, location, subscription status), purchase details (item, category, amount, season, size, color), and shopping behavior (discount applied, previous purchases, purchase frequency, review rating, shipping type, payment method)
- **Data quality:** 37 missing values in Review Rating, imputed using the median rating per product category

Tools & Technologies

Category	Tools
Data Cleaning & EDA	Python (pandas)
Database / Querying	PostgreSQL (also compatible with MySQL / SQL Server)
Visualization	Power BI
Presentation	Gamma
Report	Word / PDF

Project Steps

- 1. Data Loading** — Imported the raw dataset into a pandas DataFrame.
- 2. Exploratory Data Analysis** — Used `df.info()` and `df.describe()` to understand structure, types, and distributions.
- 3. Data Cleaning** — Checked for and imputed missing values (Review Rating filled with category-level median); standardized column names to `snake_case`.
- 4. Feature Engineering** — Created `age_group` by binning ages into quartile-based groups (Young Adult, Adult, Middle-aged, Senior); created `purchase_frequency_days` by mapping frequency labels to numeric day values; dropped `promo_code_used` after confirming redundancy with `discount_applied`.
- 5. Database Integration** — Loaded the cleaned dataset into PostgreSQL using SQLAlchemy for structured querying.
- 6. SQL Analysis** — Wrote 10 business-focused queries covering revenue by gender, discount behavior, top-rated and top-selling products, shipping comparisons, subscriber value, customer segmentation, repeat-buyer trends, and revenue by age group.
- 7. Dashboard Design** — Built an interactive Power BI dashboard with filters for subscription status, gender, category, and shipping type.
- 8. Reporting & Presentation** — Summarized findings in a written report and built a stakeholder-facing slide deck in Gamma.

Dashboard

The Power BI dashboard (`customer_behavior_dashboard.pbix`) includes:

- Key metrics: number of customers, average purchase amount, average review rating
- Subscriber vs. non-subscriber breakdown
- Revenue and sales by product category
- Revenue and sales by age group
- Interactive filters: subscription status, gender, category, shipping type

Key Results

- **Revenue by gender:** Male customers generated significantly higher total revenue than female customers.
- **Discount behavior:** 839 customers used a discount while still spending above the average purchase amount.
- **Top-rated products:** Gloves, Sandals, Boots, Hats, and Skirts had the highest average review ratings.
- **Shipping:** Express shipping customers spent slightly more on average (\$60.48) than Standard shipping customers (\$58.46).
- **Subscribers vs. non-subscribers:** Non-subscribers generated more total revenue overall, but subscribers were a smaller, high-value segment worth nurturing.

- **Discount-dependent products:** Hats, Sneakers, Coats, Sweaters, and Pants had the highest share of discounted purchases.
- **Customer segments:** 3,116 Loyal, 701 Returning, and 83 New customers — a strong existing base but a need for new-customer acquisition.
- **Repeat buyers:** Customers with more than 5 previous purchases were not strongly more likely to subscribe, suggesting subscription and loyalty are currently disconnected.
- **Revenue by age group:** Young Adults contributed the most revenue, followed closely by Middle-aged, Adult, and Senior groups.

Business Recommendations

- Promote subscription benefits to convert high-frequency shoppers into subscribers.
- Launch loyalty programs to move Returning customers into the Loyal segment.
- Reassess discount policy on high-discount-dependency products to protect margins.
- Feature top-rated and best-selling products more prominently in marketing.
- Target campaigns toward high-revenue age groups and Express-shipping users.

How to Run

1. Clone the repository

```
git clone
cd customer-shopping-behavior-analysis
```

2. Set up Python environment

```
pip install pandas sqlalchemy psycopg2-binary
```

3. Run the data cleaning notebook

Open `data_analysis_project.ipynb` in Jupyter and run all cells to load, clean, and export the dataset.

4. Load data into your SQL database

Update the database credentials in the notebook's SQLAlchemy connection step, then run it to push the cleaned data into PostgreSQL (or adapt the connection string for MySQL / SQL Server).

5. Run the SQL analysis

Execute the queries in `customer_behavior_end_to_end_Data_Analysis_PostgreSQL_project.sql` against your database to reproduce the business insights.

6. Open the dashboard

Open `customer_behavior_dashboard.pbix` in Power BI Desktop and refresh the data connection to your database (or use the saved dataset as-is).

7. View the report and presentation

- Report: Customer_Shopping_Behavior_Analysis.pdf
- Presentation: Customer-Shopping-Behavior-Analysis.pptx (built with Gamma)

Repository Structure

```
■■■ data_analysis_project.ipynb
■                                     # Python: cleaning & EDA
■■■ customer_behavior_end_to_end_Data_Analysis_PostgreSQL_project.sql
■                                     # SQL: business queries
■■■ customer_behavior_dashboard.pbix      # Power BI dashboard
■■■ Customer_Shopping_Behavior_Analysis.pdf # Project report
■■■ Customer-Shopping-Behavior-Analysis.pptx # Stakeholder presentation
■■■ README.md                            # Project documentation
```